

Myeloperoxidase Deficiency: A Comprehensive Disease Characteristics Report

Disease: Myeloperoxidase Deficiency **MONDO ID:** MONDO:0009694 | **OMIM:** #254600 | **Gene:** *MPO* (17q22.1) | **Category:** Mendelian (autosomal recessive)

Summary

Myeloperoxidase (MPO) deficiency is the **most common inherited disorder of neutrophil function**, occurring in roughly **1 in 2,000 to 4,000 individuals** in Western populations, yet it is paradoxically one of the most clinically silent primary immunodeficiencies known ([PMID: 9468285](#)). The disorder arises from biallelic loss-of-function missense mutations in the *MPO* gene (e.g., R569W, Y173C, M251T) that cause the misfolded proMPO precursor to be retained in the endoplasmic reticulum via prolonged interaction with the chaperone calnexin, followed by proteasomal degradation. The result is neutrophils and monocytes that contain precursor protein but lack mature, enzymatically active MPO, peroxidase activity, and the capacity to generate chlorinating oxidants ([PMID: 10482305](#)).

The central biological function lost in MPO deficiency is the enzyme's unique ability to use chloride as a co-substrate with hydrogen peroxide to generate **hypochlorous acid (HOCl)**, a potent antimicrobial oxidant ([PMID: 17592500](#)). Despite this, most affected individuals never develop clinical disease because the NADPH-oxidase-derived oxidant burst and other non-oxidative microbicidal systems compensate. The dominant clinical exception is **disseminated candidiasis**, which typically emerges only when a "second hit"—most classically **diabetes mellitus**—independently impairs residual antifungal killing ([PMID: 2831185](#); [PMID: 216438](#)).

This report synthesizes seven confirmed findings across the full disease-characteristics template. The overarching narrative is one of a **biochemistry-versus-clinical paradox**: a complete and readily detectable enzymatic defect that is, in isolation, biologically well-tolerated. Because MPO-derived oxidants also drive inflammatory tissue damage and vascular

disease, low-MPO states may even be partially protective against atherosclerotic cardiovascular disease and certain cancers, positioning MPO as a genuine "double-edged sword."

Key Findings

Finding 1 — MPO deficiency is the most common inherited neutrophil enzyme defect and is usually clinically silent

Hereditary MPO deficiency occurs in **1 in 2,000 to 4,000 individuals** in the general population and has traditionally been regarded as an autosomal recessive trait ([PMID: 9468285](#)). Despite this high prevalence, the overwhelming majority of affected individuals are entirely asymptomatic. The explanation is functional redundancy: NADPH-oxidase-derived reactive oxygen species and non-oxidative microbicidal systems (defensins, proteases, lactoferrin) provide sufficient host defense in the absence of MPO-generated HOCl. Clinically significant sequelae are largely restricted to a subset of patients who carry an additional predisposing condition. As the classic review states, *"In the absence of MPO, auxiliary mechanisms protect most MPO-deficient hosts from clinically significant sequelae, except for some persons with diabetes mellitus who suffer severe candidal disease"* ([PMID: 2831185](#)).

Evidence quote: *"Hereditary deficiency of MPO occurs in 1 in 2,000 to 4,000 individuals in the general population and has been generally considered an autosomal recessive trait."* ([PMID: 9468285](#))

Finding 2 — The molecular lesion: MPO missense mutations cause ER retention and defective post-translational maturation

Inherited MPO deficiency results from missense mutations in the *MPO* gene. The best-characterized alleles are **R569W**, **Y173C**, and **M251T**. In the Y173C genotype, the mutant proMPO precursor is retained in the endoplasmic reticulum through prolonged interaction with the chaperone **calnexin** and is ultimately degraded by the **20S proteasome**. Consequently, affected neutrophils contain the precursor protein but lack mature MPO subunits, peroxidase enzymatic activity, and chlorination capacity ([PMID: 10482305](#)). The R569W mutation is the most frequently encountered allele; most studied patients are **compound heterozygotes** for R569W and demonstrate a spectrum of phenotypes ranging

from complete to partial deficiency (PMID: 9468285). These mutations have served as a model system for understanding endoplasmic-reticulum quality control of secretory proteins more broadly (PMID: 15507769).

Evidence quotes: - *"In the genotype Y173C, the mutant precursor is retained in the endoplasmic reticulum by virtue of its prolonged interaction with calnexin, and it eventually undergoes degradation in the 20S proteasome."* (PMID: 10482305) - *"Most subjects were compound heterozygotes with respect to the R569W mutation and demonstrated a spectrum of phenotypes."* (PMID: 9468285)

Finding 3 — Animal models: MPO contributes to fungicidal defense, but NADPH oxidase is dominant

MPO-knockout (MPO^{-/-}) mice develop normally but exhibit **severely reduced cytotoxicity** toward *Candida albicans*, *Aspergillus fumigatus*, *Cryptococcus neoformans*, and *Klebsiella pneumoniae*, confirming that the MPO-dependent oxidative system is important for antifungal and antibacterial host defense (PMID: 15507755). Head-to-head comparison with NADPH-oxidase-deficient (X-linked chronic granulomatous disease, X-CGD) mice reveals a clear hierarchy: X-CGD mice suffer shorter survival and 10–100× higher fungal tissue burdens than MPO^{-/-} mice. Critically, MPO cannot function without NADPH-oxidase-derived hydrogen peroxide, which is its obligate substrate. **However**, at the highest *Candida* inocula, the mortality of MPO^{-/-} mice approached that of CGD mice, indicating that MPO becomes rate-limiting under a heavy pathogen load (PMID: 12521119; PMID: 16940954).

Evidence quotes: - *"Both MPO-deficient (MPO^{-/-}) and NADPH-oxidase-deficient (X-linked chronic granulomatous disease [X-CGD]) mice showed increased susceptibility to pulmonary infections with *Candida albicans* and *Aspergillus fumigatus* compared with normal mice, and the X-CGD mice exhibited shorter survivals than MPO^{-/-} mice."* (PMID: 12521119) - *"MPO is unable to play a role in host defense in the absence of NADPH-oxidase."* (PMID: 12521119)

Finding 4 — The core enzymatic reaction: $\text{H}_2\text{O}_2 + \text{Cl}^- \rightarrow \text{HOCl}$, a double-edged sword

MPO is a member of the heme peroxidase-cyclooxygenase superfamily and is abundantly stored in the azurophilic (primary) granules of neutrophils. Its **unique catalytic activity** is the use of chloride as a co-substrate with hydrogen peroxide to generate **hypochlorous acid (HOCl)**, a potent antimicrobial agent (PMID: 17592500). This same chemistry is a liability: MPO-derived oxidants contribute to host tissue damage and to the initiation and propagation of

acute and chronic vascular inflammatory disease, and circulating MPO levels predict adverse cardiac events. HOCl oxidatively modifies proteins—through amino-acid side-chain modification, backbone fragmentation, and aggregation—driving chronic inflammatory pathology (PMID: 31867603). This dual nature is the mechanistic foundation for why loss of MPO can be biologically tolerated and even confer protection in some disease contexts.

Evidence quotes: - *"A unique activity of MPO is its ability to use chloride as a cosubstrate with hydrogen peroxide to generate chlorinating oxidants such as hypochlorous acid, a potent antimicrobial agent."* (PMID: 17592500) - *"MPO-derived oxidants contribute to tissue damage and the initiation and propagation of acute and chronic vascular inflammatory disease."* (PMID: 17592500)

Finding 5 — Classification and diagnosis: a primary immunodeficiency detected incidentally by hematology analyzers

MPO deficiency is listed among the primary immunodeficiencies that predispose to fungal infection (PMID: 17551753). Its high apparent prevalence is itself an artifact of modern laboratory medicine: *"the relatively high prevalence of inherited MPO deficiency was an unanticipated insight provided by the widespread use of automated flow cytometry for the enumeration of leukocytes in clinical specimens"* (PMID: 10482305). Automated hematology analyzers use the peroxidase (MPO) channel to perform leukocyte differentials, so MPO-deficient neutrophils are flagged incidentally in otherwise healthy people. Confirmation relies on **cytochemical peroxidase staining** of blood smears—MPO-deficient neutrophils and monocytes are peroxidase-negative while eosinophils remain positive (via eosinophil peroxidase, EPO)—and on direct MPO enzyme-activity assays. A known diagnostic pitfall is that EPO, which is normally expressed in MPO-deficient subjects, can confound leukocyte peroxidase measurements through eosinophil contamination (PMID: 9468285).

Evidence quotes: - *"the relatively high prevalence of inherited MPO deficiency was an unanticipated insight provided by the widespread use of automated flow cytometry for the enumeration of leukocytes in clinical specimens."* (PMID: 10482305) - *"Eosinophil peroxidase (EPO) also contributes to the peroxidase activity of blood leukocytes. Because EPO expression is normal in MPO-deficient subjects, eosinophil contamination can significantly contribute to peroxidase activity."* (PMID: 9468285)

Finding 6 — The MPO trade-off: low-expression -463A allele protects the heart but a high-expression genotype raises cancer risk

Distinct from the rare loss-of-function missense mutations that cause hereditary deficiency, a **common functional promoter polymorphism, -463 G>A (rs2333227)**, quantitatively modifies MPO expression: the G allele confers higher expression than the A allele (PMID: 11479475). In French-Canadians, the low-expression **AA genotype was associated with markedly decreased coronary artery disease (CAD) risk** (recessive model OR 0.138, 95% CI 0.040–0.474), and carriage of the A allele (AA/AG vs GG) was protective (OR 0.639, 95% CI 0.436–0.937). Conversely, the high-expression **GG genotype hastened hepatocellular carcinoma** in HCV-related cirrhosis (HR 2.8, 95% CI 1.7–4.4; PMID: 21907168), and higher MPO activity has been implicated in leukemia and lung cancer through metabolic activation of carcinogens such as benzene (PMID: 17479404). These associations extend the biological "double-edged sword" to the population-genetics level, though some studies find no relationship between the promoter polymorphisms and neutrophil MPO release or cardiovascular risk (PMID: 19877306), so the effect is modest and not fully reproducible.

Evidence quotes: - *"In a recessive model patients with the AA genotype had a decreased risk of CAD (odds ratio 0.138, 95% confidence interval 0.040-0.474)."* (PMID: 11479475) - *"the G allele associated with a higher level of MPO expression than the A allele."* (PMID: 11479475) - *"HCC occurrence was increased in patients with either the homozygous GG-MPO genotype (HR=2.8 [1.7-4.4])."* (PMID: 21907168)

Finding 7 — The clinical phenotype is unmasked by a second hit: diabetes mellitus converts silent deficiency into candidal susceptibility

The classic clinical scenario for symptomatic MPO deficiency is a **diabetic patient with disseminated or invasive candidiasis**. In a landmark case, an MPO-deficient diabetic patient's granulocytes showed normal phagocytosis but microbicidal activity that was *"almost nil with regard to Candida albicans"*; crucially, *"Fungicidal activity of normal granulocytes was shown to be impaired during the in vitro artificial hyperglycemic condition"*—demonstrating that hyperglycemia independently degrades residual killing (PMID: 216438). MPO-deficient diabetics have developed *Candida albicans* liver abscess (PMID: 217268; PMID: 199939). Even in non-diabetics, disseminated pustular candidal dermatitis has occurred, particularly under limited-spectrum antibiotic therapy (PMID: 9114158). The unifying principle from Nauseef is that clinically significant disease is limited *"except for some persons with diabetes mellitus who suffer severe candidal disease"* (PMID: 2831185).

Evidence quotes: - "Fungicidal activity of normal granulocytes was shown to be impaired during the *in vitro* artificial hyperglycemic condition." (PMID: 216438) - "Patients who develop rapidly disseminated fungal dermatitis while they are receiving antimicrobial therapy that is relatively limited in coverage should be evaluated for myeloperoxidase deficiency." (PMID: 9114158)

Comprehensive Section-by-Section Report

1. Disease Information

Myeloperoxidase deficiency is an inherited disorder in which neutrophils and monocytes lack functional myeloperoxidase, the azurophilic-granule heme enzyme responsible for generating hypochlorous acid during the oxidative burst. It is the most common inherited defect of neutrophils. Two forms exist: **complete (total) deficiency** and **partial deficiency**, reflecting the genotype (homozygous vs. compound heterozygous / heterozygous for hypomorphic alleles).

Key identifiers:

Resource	Identifier
MONDO	MONDO:0009694
OMIM (phenotype)	#254600
OMIM (gene)	*606989
Gene (HGNC)	<i>MPO</i> , HGNC:7218
Gene locus	17q22.1
UniProt (protein)	P05164 (PERM_HUMAN)
Orphanet	ORPHA:59181
ICD-10	D70/D72.0 (functional disorders of neutrophils)
MeSH	related to "Peroxidase" / "Leukocyte Disorders"

Synonyms / alternative names: MPO deficiency; myeloperoxidase deficiency; hereditary myeloperoxidase deficiency; leukocyte myeloperoxidase deficiency; total/partial myeloperoxidase deficiency; Peroxidase deficiency.

Source of information: The knowledge base for this disorder is derived largely from **aggregated disease-level resources** (OMIM, primary case series, and reviews) combined with individual patient case reports; the high prevalence estimate itself derives from population-scale automated hematology analyzer data ([PMID: 10482305](#)).

2. Etiology

Causal factors: The disease is **genetic (Mendelian, autosomal recessive)**, caused by biallelic loss-of-function mutations in *MPO*. It is not infectious or environmental in origin, though environmental/clinical co-factors (see below) determine whether it is expressed clinically.

Genetic risk factors: Causal variants include the missense mutations **R569W** (the most common, [PMID: 9468285](#)), **Y173C**, and **M251T** ([PMID: 10482305](#), [PMID: 15507769](#)). The common promoter polymorphism **-463 G>A (rs2333227)** is a quantitative expression modifier rather than a cause of deficiency ([PMID: 11479475](#)).

Environmental/clinical risk factors for symptomatic disease (second hits): Diabetes mellitus / hyperglycemia is the principal unmasking factor, as it independently impairs granulocyte fungicidal activity ([PMID: 216438](#)). Additional triggers include limited-spectrum antimicrobial therapy (allowing fungal overgrowth) ([PMID: 9114158](#)) and high pathogen inoculum ([PMID: 12521119](#)).

Protective factors: From the host-defense standpoint, an intact **NADPH oxidase** and non-oxidative microbicidal systems are the key compensatory/protective mechanisms ([PMID: 12521119](#)). From a cardiovascular standpoint, the **low-expression -463A allele** may be protective against CAD ([PMID: 11479475](#)).

Gene–environment interaction: The paradigmatic GxE interaction is **MPO genotype × diabetes mellitus**: neither alone produces disseminated candidiasis, but their combination does ([PMID: 216438](#); [PMID: 217268](#)). A second GxE axis is **MPO-463 genotype × carcinogen exposure** (e.g., benzene, tobacco smoke), where higher MPO activity enhances procarcinogen activation ([PMID: 17479404](#)).

3. Phenotypes

Most individuals are **asymptomatic** (laboratory abnormality only). When present, phenotypes are infectious.

Phenotype	Type	Onset	Severity	Frequency	Suggested HPO
Peroxidase-negative neutrophils/monocytes	Laboratory abnormality	Congenital	—	~100% of affected	HP:0011990 (abnormal granulocyte morphology) / lab finding
Recurrent/disseminated candidiasis	Clinical sign / infection	Any age (typically adult)	Severe when present	Rare; mainly with diabetes	HP:0002728 (chronic mucocutaneous candidiasis)
Candidal (hepatic) abscess	Clinical sign	Adult	Severe	Rare	HP:0100523 (hepatic abscess)
Pustular candidal dermatitis	Physical manifestation	Variable	Moderate–severe	Rare	HP:0200037 (pustule)
Increased susceptibility to fungal/bacterial infection	Clinical sign	Variable	Mild–severe	Minority	HP:0002719 (recurrent infections)

Quality-of-life impact: For the asymptomatic majority, there is **no measurable QoL impact**; the condition is compatible with normal life. For the rare symptomatic subset, invasive candidiasis carries substantial morbidity and potential mortality. No disease-specific EQ-5D/SF-36 data are available for MPO deficiency.

4. Genetic / Molecular Information

Causal gene: *MPO* (myeloperoxidase), chromosome **17q22.1**, OMIM gene 606989; disease OMIM #254600. Protein: myeloperoxidase, UniProt **P05164**, a heme peroxidase-cyclooxygenase superfamily enzyme ([PMID: 17592500](https://pubmed.ncbi.nlm.nih.gov/17592500/)).

Pathogenic variants:

Variant	Type	Consequence	Notes
R569W	Missense	Loss of function; defective maturation	Most common allele; often compound heterozygous (PMID: 9468285)
Y173C	Missense	ER retention via calnexin; 20S proteasomal degradation	Model for ER quality control (PMID: 10482305)
M251T	Missense	Loss of function / defective maturation	(PMID: 15507769)
-463 G>A (rs2333227)	Promoter SNP	Expression modifier (not causal)	Common quantitative variant (PMID: 11479475)

Functional consequence: Loss of function — absence of mature MPO subunits, peroxidase activity, and chlorination capacity (PMID: 10482305). **Origin:** germline. **Modifier genes:** *CYBB*/NADPH-oxidase components determine the functional consequence of MPO loss (compensatory); the *MPO* -463 promoter genotype modifies expression quantitatively. **Epigenetic / chromosomal abnormalities:** none characteristically associated with the Mendelian form.

5. Environmental Information

The disorder is genetically determined; environmental factors act only as **modifiers of clinical expression**. Relevant factors: **hyperglycemia/diabetes** (impairs residual fungicidal activity, PMID: 216438); **broad- vs. limited-spectrum antibiotic exposure** (fungal overgrowth risk, PMID: 9114158); and **carcinogen exposure** interacting with MPO-463 genotype (PMID: 17479404). **Infectious agents** are consequences, not causes: chiefly *Candida albicans*, and in models *Aspergillus fumigatus*, *Cryptococcus neoformans*, and *Klebsiella pneumoniae* (PMID: 15507755).

6. Mechanism / Pathophysiology

Ordered causal chain (from genetic lesion to clinical manifestation):

1. Biallelic MPO missense mutation (e.g., R569W, Y173C, M251T)
 - | leads to
2. Misfolded proMPO precursor
 - | leads to (Y173C) prolonged calnexin binding → ER retention
3. Proteasomal (20S) degradation of the precursor
 - | results in
4. Absence of mature, enzymatically active MPO in azurophilic granules
 - | results in
5. Loss of the reaction $\text{H}_2\text{O}_2 + \text{Cl}^- \rightarrow \text{HOCl}$ (hypochlorous acid)
 - | results in
6. Impaired oxidative (HOCl-mediated) microbial killing
 - BRANCH A (usual outcome): NADPH oxidase + non-oxidative systems compensate → CLINICALLY SILENT
 - BRANCH B (second hit present, e.g., diabetes/hyperglycemia, or heavy pathogen inoculum): residual killing overwhelmed
 - | leads to
 - 7. Failure to contain *Candida albicans*
 - | leads to
 - 8. Disseminated / invasive candidiasis (e.g., hepatic abscess, pustular dermatitis) → CLINICAL DISEASE

A parallel, **beneficial branch** stems from step 5: because HOCl also damages host tissue and oxidatively modifies host proteins ([PMID: 31867603](#)), reduced MPO oxidant output *lowers* vascular inflammatory injury, plausibly underlying the reduced CAD risk seen with the low-expression -463A allele ([PMID: 11479475](#)).

Molecular pathways / biochemistry: The lost reaction is the halide-oxidation cycle of a heme peroxidase (Compound I / Compound II redox intermediates); MPO uniquely oxidizes chloride to HOCl ([PMID: 17592500](#)). **Cellular processes:** neutrophil respiratory burst, phagolysosomal microbial killing, and inflammation. **Protein dysfunction:** misfolding → ER retention → proteasomal degradation (loss of function) ([PMID: 10482305](#)). **Immune involvement:** primary immunodeficiency of the innate/phagocytic arm ([PMID: 17551753](#)). **Tissue-damage mechanism (the flip side):** HOCl-mediated oxidative protein modification, backbone fragmentation, and aggregation in chronic inflammation ([PMID: 31867603](#)).

Suggested ontology terms: GO:0006979 (response to oxidative stress); GO:0042744 (hydrogen peroxide catabolic process); GO:0002446 (neutrophil mediated immunity); GO:0043312 (neutrophil degranulation). Chemicals: CHEBI:24757 (hypochlorous acid); CHEBI:16240 (hydrogen peroxide); CHEBI:17996 (chloride). Cell types: CL:0000775 (neutrophil), CL:0000576 (monocyte). Upstream = mutation/ER retention; downstream = impaired killing and infection.

7. Anatomical Structures Affected

- **Primary cells:** neutrophil granulocytes (CL:0000775) and monocytes (CL:0000576); eosinophils are spared (they express eosinophil peroxidase) (PMID: 217268).
- **Subcellular compartment:** azurophilic (primary) granules — GO:0042582 (azurophil granule); the enzymatic block occurs in the **endoplasmic reticulum** (GO:0005783) (PMID: 10482305).
- **Body system:** hematopoietic / innate immune system (UBERON:0002390 hematopoietic system; UBERON:0000178 blood).
- **Secondary organ involvement (only in symptomatic disease):** liver (candidal abscess; UBERON:0002107), skin (pustular candidal dermatitis; UBERON:0002097), and lungs in animal models (UBERON:0002048). Involvement is typically systemic/disseminated rather than lateralized.

8. Temporal Development

- **Onset:** the enzymatic defect is **congenital**; the laboratory abnormality is present from birth. Clinical infection, when it occurs, is usually **adult-onset** and tied to acquisition of a second hit (e.g., diabetes).
- **Course:** the underlying deficiency is **chronic and lifelong** but static. Infectious episodes are **episodic/acute**, precipitated by co-factors.
- **Progression:** the enzyme defect itself does not progress. There are no defined disease "stages." **Critical period for intervention:** aggressive antifungal therapy at the onset of disseminated candidiasis; and control of comorbid hyperglycemia as ongoing prophylaxis.

9. Inheritance and Population

- **Prevalence:** approximately **1 in 2,000 to 4,000** ($\approx 25\text{--}50$ per 100,000) for at least partial deficiency in Western populations; complete deficiency is rarer (PMID: 9468285).
- **Inheritance:** **autosomal recessive** with **variable expressivity** (PMID: 9468285; PMID: 199939). Compound heterozygosity (especially involving R569W) is common.
- **Penetrance:** biochemically high (the enzyme defect is consistently detectable), but **clinical penetrance is very low** — most homozygotes/compound heterozygotes are asymptomatic.
- **Expressivity:** variable, from partial to complete enzyme loss and from silent to symptomatic (PMID: 199939).

- **Sex ratio / geography / founder effects:** no strong sex predilection or well-defined founder effect is established for the classic missense alleles; the -463 promoter allele frequencies vary by population ([PMID: 17479404](#)).

10. Diagnostics

- **Incidental detection:** flagged by **automated hematology analyzers** using the peroxidase (MPO) channel during routine leukocyte differentials ([PMID: 10482305](#)).
- **Confirmatory tests: cytochemical peroxidase staining** of a peripheral blood smear (neutrophils/monocytes peroxidase-negative, eosinophils positive) and **MPO enzyme-activity assays** ([PMID: 9468285](#); [PMID: 217268](#)).
- **Diagnostic pitfall:** eosinophil peroxidase (EPO), normally expressed in MPO-deficient subjects, can contribute to measured leukocyte peroxidase and confound results ([PMID: 9468285](#)).
- **Distinguishing from CGD:** flow-cytometric assays of the oxidative burst (e.g., dihydrorhodamine-123) are normal in MPO deficiency but abnormal in chronic granulomatous disease—an important differential ([PMID: 7813334](#)).
- **Genetic testing:** *MPO* single-gene sequencing / immunodeficiency gene panels can confirm biallelic variants; useful for family counseling but not required for diagnosis.
- **Differential diagnosis:** chronic granulomatous disease (NADPH-oxidase defect), other neutrophil functional disorders, and secondary/acquired MPO deficiency (e.g., myelodysplasia, certain drugs).

11. Outcome / Prognosis

- **Overall prognosis is excellent.** Most individuals have **normal life expectancy** and no increased infection burden.
- **Mortality/morbidity** is confined to the rare symptomatic subset with invasive candidiasis, where outcome depends on prompt antifungal therapy and control of comorbid conditions ([PMID: 2831185](#); [PMID: 217268](#)).
- **Prognostic factors:** presence of diabetes mellitus/hyperglycemia, completeness of enzyme deficiency, pathogen inoculum, and breadth of antimicrobial coverage ([PMID: 216438](#); [PMID: 12521119](#)).
- **Potential benefit:** reduced MPO oxidant load may lower atherosclerotic cardiovascular risk ([PMID: 11479475](#)), an unusual "protective" dimension of an immunodeficiency.

12. Treatment

- **No specific therapy** exists or is required for the enzyme deficiency itself.
- **Management is directed at infections and comorbidities:** prompt **antifungal therapy** for candidiasis (e.g., azoles, echinocandins, lipid amphotericin B formulations, which have expanded options for fungal infection in primary immunodeficiencies, [PMID: 17551753](#)) and **tight glycemic control** in diabetic patients as the single most important preventive measure ([PMID: 216438](#)).
- **Antibiotic stewardship:** avoiding unnecessary limited-spectrum regimens that promote fungal overgrowth ([PMID: 9114158](#)).
- **Advanced/gene therapy:** none developed or indicated given the benign natural history. **NCIT suggestions:** NCIT:C305 (Amphotericin B), NCIT:C1471 (antifungal agent), NCIT:C1505 (azole antifungal).

13. Prevention

- **Primary prevention:** not applicable to the genetic defect; **genetic counseling** for affected families (autosomal recessive recurrence risk) is appropriate.
- **Secondary/tertiary prevention:** the highest-yield intervention is **prevention and control of diabetes/hyperglycemia** to avoid unmasking clinical disease ([PMID: 216438](#)); clinical vigilance for candidal infection in at-risk deficient patients; and judicious antibiotic use ([PMID: 9114158](#)).
- **Screening:** no population-based newborn screening is indicated because clinical penetrance is very low; incidental analyzer detection effectively serves as opportunistic case-finding.

14. Other Species / Natural Disease

- **Model species with MPO orthologs:** *Mus musculus* (mouse *Mpo*), used in knockout studies ([PMID: 15507755](#)). MPO is evolutionarily conserved within the heme peroxidase-cyclooxygenase superfamily ([PMID: 17592500](#)).
- **Comparative biology:** the mouse MPO^{-/-} phenotype (impaired fungicidal activity, normal development) parallels the human condition, supporting conserved mechanism ([PMID: 12521119](#)). No prominent naturally occurring companion-animal MPO deficiency disorder is documented in the reviewed literature. Not zoonotic.

15. Model Organisms

- **Primary model:** MPO-knockout (MPO^{-/-}) mouse ([PMID: 15507755](#)).
- **Comparator model:** X-linked CGD (NADPH-oxidase-deficient) mouse, and MPO^{-/-} × X-CGD double knockouts, used to dissect the relative contributions of MPO vs. the oxidase ([PMID: 12521119](#)).
- **Phenotype recapitulation:** good — MPO^{-/-} mice develop normally but show reduced antifungal killing, matching the human "silent-except-under-stress" phenotype; at high inoculum they approach CGD-level mortality, mirroring the human second-hit phenomenon ([PMID: 15507755](#); [PMID: 16940954](#)).
- **Limitations:** murine neutrophil biology differs from human; models do not fully capture the diabetes-interaction axis or human population genetics of the -463 polymorphism.
- **In vitro systems:** patient granulocyte functional assays and neutrophil lysate MPO-activity assays have been central to mechanistic work ([PMID: 216438](#); [PMID: 19877306](#)).

Mechanistic Model / Interpretation

MPO deficiency is best understood as a **conditional, redundancy-buffered immunodeficiency**. The genetic lesion reliably abolishes a specific biochemical capability—HOCl generation—yet the clinical system it feeds into is robust because a parallel, more powerful effector (NADPH oxidase) and non-oxidative killing remain intact. This explains the disease's defining paradox: a 100%-penetrant biochemical defect with near-0% clinical penetrance.

The "**double-edged sword**" framing unifies the seemingly disparate findings:

Context	MPO/HOCI HIGH	MPO/HOCI LOW (deficiency or -463A)
Antifungal defense	Effective killing	Impaired (Findings 3, 4)
Under a second hit (diabetes, heavy inoculum)	Protected	Susceptible → candidiasis (Finding 7)
Vascular inflammation / CAD	Higher risk	Reduced risk (Finding 6)
Carcinogen activation (leukemia, HCC)	Higher risk (GG genotype)	Reduced risk (Finding 6)

The clinical decision node is the **second hit**. Absent one, the deficiency is a laboratory curiosity; present one, it can produce life-threatening invasive fungal disease. Diabetes is the archetype because hyperglycemia independently degrades the residual killing capacity that would otherwise compensate.

Evidence Base

PMID	Title (abbrev.)	Role in this report
9468285	<i>Inheritance & R569W mutation</i>	Prevalence, AR inheritance, compound heterozygosity, EPO diagnostic pitfall
10482305	<i>ER quality control, MPO deficiency</i>	ER retention/calnexin/proteasome mechanism; incidental analyzer diagnosis
15507769	<i>Structural features from MPO deficiency</i>	Catalog of causal missense mutations
15507755	<i>In vivo role of MPO</i>	MPO-KO fungicidal defect
12521119	<i>MPO vs NADPH-oxidase</i>	Relative contributions; MPO depends on oxidase
16940954	<i>MPO in antifungal defense (review)</i>	High-inoculum equivalence with CGD
17592500	<i>MPO: drug target?</i>	Core HOCl chemistry; tissue-damage duality
31867603	<i>HOCl protein modification</i>	Tissue-damage mechanism
11479475	<i>MPO -463 & CAD (French-Canadians)</i>	Protective low-expression allele
21907168	<i>MPO promoter & HCC</i>	High-expression genotype raises cancer risk
17479404	<i>MPO SNPs & leukemia</i>	Carcinogen-activation trade-off
216438	<i>Hereditary MPO deficiency (case)</i>	Hyperglycemia impairs fungicidal activity
217268	<i>MPO deficiency + diabetes + Candida liver abscess</i>	Second-hit clinical syndrome
9114158	<i>Pustular candidal dermatitis</i>	Non-diabetic trigger; management advice
2831185	<i>MPO deficiency (review)</i>	Clinical silence except diabetic candidiasis
17551753	<i>Fungal infections in PIDs</i>	Classifies MPO deficiency among PIDs
7813334	<i>Flow cytometry for CGD variants</i>	DHR assay distinguishes MPO deficiency from CGD
19877306	<i>MPO promoter SNPs & neutrophil activation</i>	Challenges reproducibility of -463 CVD link
199939	<i>Hereditary MPO deficiency (genetics)</i>	AR transmission, variable expressivity

Challenging/qualifying evidence: [PMID: 19877306](#) found no relationship between the -129/-463 promoter polymorphisms and neutrophil MPO release or reactive oxygen production, cautioning that the -463/CVD association is modest and inconsistently reproducible. This tempers Finding 6.

Limitations and Knowledge Gaps

1. **No primary dataset was analyzed** — this is a literature-synthesis report; findings rest on published case series, reviews, and mouse models rather than newly generated statistics.
 2. **Prevalence uncertainty** — the 1:2,000–4,000 figure is Western/analyzer-derived and may not generalize globally; complete vs. partial deficiency proportions are imprecise.
 3. **Clinical penetrance is not precisely quantified** — the fraction of deficient individuals who ever develop candidiasis, and its dependence on diabetes severity, lacks large cohort estimates.
 4. **-463 polymorphism associations are inconsistent** — cardiovascular and cancer associations vary across populations and are not robustly reproducible ([PMID: 19877306](#)).
 5. **Genotype–phenotype granularity** — beyond R569W/Y173C/M251T, the full variant spectrum, allele frequencies (gnomAD), and ACMG classifications were not exhaustively enumerated here.
 6. **Human GxE quantification** — the diabetes × MPO interaction is well-supported mechanistically but not quantified with modern epidemiology.
-

Proposed Follow-up Experiments / Actions

1. **Quantify clinical penetrance** via a registry/EHR cohort of MPO-deficient individuals stratified by diabetes status, estimating the absolute and relative risk of invasive candidiasis.
2. **Comprehensive variant curation** — mine ClinVar/gnomAD for the full *MPO* pathogenic variant spectrum, allele frequencies, and ACMG/AMP classifications; report carrier frequencies by ancestry.

3. **Meta-analysis of the -463 G>A polymorphism** across cardiovascular and oncologic endpoints to resolve the reproducibility conflict between [PMID: 11479475](#) and [PMID: 19877306](#).
 4. **Mechanistic modeling of the second hit** — in vitro MPO-deficient neutrophils under controlled glucose to dissect how hyperglycemia degrades residual (NADPH-oxidase-dependent) *Candida* killing.
 5. **Test the cardiovascular-protection hypothesis prospectively** in genetically confirmed complete MPO-deficient individuals (atherosclerosis imaging vs. matched controls).
 6. **HPO/ontology annotation package** — formalize the phenotype-frequency table above into HP-term associations for the disease knowledge base, with CL:0000775 (neutrophil), UBERON, and CHEBI:24757 (hypochlorous acid) links.
-

Report compiled from a 5-iteration autonomous investigation; 7 confirmed findings; 28 papers reviewed. Evidence source types span human clinical case series/reviews, mouse knockout models, and in vitro neutrophil assays.

Generated by OpenScientist — Scientific Hypothesis Agent for Novel Discovery